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Purification of the Transfer Enzymes from Reticulocytes and Properties of the Transfer Reaction[1]

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The transfer of ^{14}C-amino acids from ^{14}C-aminoacyl-RNA into globin polypeptides on reticulocyte ribosomes required guanosine 5′-triphosphate (GTP) and two soluble enzyme fractions. The detailed purification of these transfer enzymes is described. The transfer reaction occurred at concentrations of $MgCl_2$ and KCl which were optimal for cell-free synthesis of hemoglobin. Other nucleoside triphosphates could not substitute for GTP. About 70% of the radioactivity of the ribosome-bound product was found in acid-precipitable polypeptides. The remaining 30% was found in an acid-soluble fraction containing mostly small peptides. A comparison suggests that amino acids are added on to existing polypeptide chains on the ribosomes in this hemoglobin transfer system, whereas chain initiation is the predominant reaction in a transfer system stimulated with polyuridylic acid.

Several laboratories have reported that guanosine 5′-triphosphate (GTP) and more than one enzyme fraction are involved in the transfer of amino acids from amino acyl-RNA to ribosome-bound polypeptides (1). Our work using a cell-free system from rabbit reticulocytes has revealed that two soluble enzyme fractions and GTP are needed for the formation of small polypeptides of phenylalanine from phenylalanyl-RNA in the presence of poly U (polyuridylic acid) (2). In a preliminary note we have reported that these same enzyme fractions are necessary for the transfer of leucine-^{14}C from leucyl-^{14}C RNA into polypeptides using a reticulocyte system in the absence of poly U (3). This transfer system was shown previously to synthesize soluble hemoglobin under optimal conditions (4). Here we describe in detail the methods used in isolation of the purified enzyme fractions and the general properties of this hemoglobin transfer system.

To Richard S. Schweet, pioneer in the field of peptide synthesis—in memoriam.

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EXPERIMENTAL METHODS

Preparation of Enzyme Fractions

(*a*) $(NH_4)_2SO_4$ *fractionation.* Rabbits were made anemic with phenylhydrazine, and the reticulocytes were collected and lysed as previously described (5). Figure 1 shows a general scheme for the purification of the enzyme fractions. The solution from the 68,000*g* centrifugation of the reticulocyte lysate was stored at −15° until used. About 1700 ml of this supernatant fraction were thawed overnight at 5°, and neutralized protamine sulfate (10 mg/ml) was added to a final

concentration of 0.34 mg/ml. The suspension was stirred for 30 minutes at 0°. All subsequent operations were performed at 0–4°. The protamine sulfate precipitate was removed by centrifugation for 20 minutes at 10,000*g*. The supernatant fraction was made 0.1 M in Tris-HCl by the addition of a sufficient volume of 2 M Tris-HCl, pH 7.5, at 23°. Then 1 M acetic acid was added slowly to bring the pH to 6.5.

The solution was brought to 40% saturation in $(NH_4)_2SO_4$ by the slow addition of 22.6 gm of finely powdered $(NH_4)_2SO_4$ (enzyme grade, Mann Research Laboratories, Inc.) per 100 ml. The suspension was stirred gently for 30 minutes, and the precipitate was removed by centrifugation at 10,000*g* for 20 minutes. The supernatant fraction was brought to 70% saturation in $(NH_4)_2SO_4$ by the addition of 21.0 gm/100 ml of the original pH 6.5 solution. After stirring for 30 minutes, the suspension was centrifuged, and the supernatant phase was discarded. The precipitate was dissolved in 850 ml of 0.10 M Tris-HCl, pH 6.5 at 0°, containing 1 mM glutathione (GSH) and 0.10 mM sodium EDTA. The resulting solution was brought to approximately 70% saturation in $(NH_4)_2SO_4$ by the addition of 370 gm, and the precipitate was collected by centrifugation. The precipitate was dissolved in a minimal volume of 0.02 M Tris-HCl, pH 7.5, at 23°, containing 0.5 mM GSH and 0.1 mM EDTA, and the solution was dialyzed overnight against the same buffer. The dialyzed solution was clarified by centrifugation (10,000*g* for 10 minutes) and made 1 mM in EDTA by the addition of a sufficient volume of 0.2 M EDTA. About 4 gm of protein were obtained in 60–75 ml. At this stage the enzyme preparation (ammonium sulfate fraction) was stable enough to be stored for at least 3 months at −20°.

(*b*) *Fractionation on calcium phosphate gel.* The above $(NH_4)_2SO_4$ fraction was diluted to 20 mg of protein per milliliter and simultaneously made 0.10 M in Tris-HCl by the addition of sufficient 2 M Tris-HCl, pH 7.5. Then 0.4 volume (per volume of diluted enzyme) of a gel suspension containing 30 mg of calcium phosphate per milliliter (6) was added dropwise with stirring. The mixture was stirred for 45 minutes at 0°. The gel was collected by centrifugation at 5000*g* for 10 minutes. The supernatant phase (gel fraction S-1) was discarded. The gel was washed by gentle homogenization in 1.2 volumes (per volume of original diluted enzyme) of 0.075 M Tris-HCl, pH 7.5, and reisolated by centrifugation. The supernatant wash was discarded.

The gel was homogenized in 0.4 volume of 0.1 M phosphate (K^+) buffer, pH 7.0 ,and the suspension was stirred for 15 minutes. After centrifugation the resulting supernatant phase (gel fraction S-2) was made 0.01 M in mercaptoethylamine and dialyzed overnight against 0.002 M Tris-HCl, pH 7.5, containing 0.01 M mercaptoethylamine and 0.001 M EDTA. The gel was washed by homogenization in 2.4 volumes of the 0.1 M phosphate buffer, pH 7.0. The mixture was centrifuged, and the wash was discarded. The gel was homogenized finally in 0.4 volume of 0.3 M phosphate (K^+) buffer, pH 7.5. The suspension was stirred for 15 minutes, and the gel was removed by centrifugation and discarded. The supernatant phase (gel fraction S-3) was dialyzed overnight against 0.002 M Tris-HCl, pH 7.5. These dialyzed enzyme fractions S-2 and S-3 were not stored frozen. They were purified further on DEAE(diethylaminoethyl)-cellulose columns as described below.

(*c*) *DEAE-Cellulose purification.* The DEAE-cellulose (Biorad Laboratories) was washed in a batch by suspension and decantation with water, 0.5 N HCl, and 0.5 N NaOH, in succession. The cellulose was washed finally several times with water, and the pH of the slurry was brought to 7.0 with 1 N HCl. The cellulose was washed again with water and equilibrated for storage with 0.002 M Tris-HCl, pH 7.5.

The dialyzed fraction S-2 (approximately 400 mg protein) was applied to a 2 × 35-cm column of DEAE-cellulose previously equilibrated with a starting buffer of 0.002 M Tris-HCl, pH 7.5, at 23°, containing 0.01 M mercaptoethylamine and 1.0 mM magnesium ammonium EDTA (7). The column was washed with 200 ml of the starting buffer, at a flow rate of 60 ml per hour. Some contaminating proteins were eluted at this step. The transfer fraction (TF-2) with the peptide synthetase activity (2) was eluted then in a broad band with 200 ml of 0.10 M NaCl in the starting buffer. Fractions of 10 ml were collected. Fractions of the band having an absorbance at 280 mμ greater than 3.0 were pooled and dialyzed overnight against 0.02 M Tris-HCl, pH 7.5, containing 0.5 mM GSH and 0.1 mM magnesium ammonium EDTA. This peptide synthetase (TF-2) enzyme (approximately 60 mg protein) was made 0.025 M in GSH and 0.001 M in magnesium ammonium EDTA and stored at −20°.

The dialyzed gel fraction S-3 (approximately 300 mg protein) was applied to a 2 × 35-cm column of DEAE-cellulose previously equilibrated with 0.002 M Tris-HCl, pH 7.5, at 23°. Contaminating proteins were eluted in succession with 200 ml of 0.002 M Tris-HCl, pH 7.5, and 200 ml of 0.10 M NaCl in the same buffer. The transfer fraction (TF-1) with the binding enzyme activity (2) was eluted with 200 ml of 0.25 M NaCl in the Tris buffer, at a flow rate of 60 ml per hour. Fractions (10 ml each) of the protein band having an absorbance (280 mμ) greater than 3.0 were pooled and dialyzed

against 0.02 M Tris-HCl, pH 7.5. This binding enzyme (TF-1) fraction contained approximately 100 mg of protein and was stored at −20°.

The conditions for stability upon storage of the enzyme fractions TF-1 and TF-2 were different. The binding enzyme could be stored in 0.02 M Tris-HCl, pH 7.5, for several months at −20° without loss in transfer activity. The peptide synthetase was very labile when stored using these conditions (see RESULTS). However, the peptide synthetase was stable when 0.025 M GSH and 0.001 M magnesium ammonium EDTA were included in the buffer.

PREPARATION OF KCl-WASHED RIBOSOMES

A ribosome fraction was isolated from the reticulocyte lysate by centrifugation at 78,000*g* for 90 minutes. The ribosome pellet was homogenized in a small volume of 0.25 M sucrose, and aggregated material was removed by centrifugation at 10,000*g* for 10 minutes. The solution of ribosomes was diluted with 0.10 M KCl in cold medium B (0.25 M sucrose, 0.0175 M $KHCO_3$, 0.002 M $MgCl_2$) to a volume two-thirds that of the original cell lysate. The ribosomes were reisolated by centrifugation, and the pellets were homogenized and pooled in a small volume of 0.25 M sucrose. After removal of the aggregates by centrifugation, the absorbance at 260 mμ of the solution was determined. The solution was diluted with 0.25 M sucrose to 15 mg of ribosomes per milliliter, assuming 1 mg of ribosomes per milliliter has an absorbance of 11.2. These KCl-washed ribosomes were stored frozen at −60° in small samples.

PREPARATION OF AMINOACYL-^{14}C RNA

Radioactive aminoacyl-RNA was prepared with the pH 5.0 enzyme fraction from guinea pig liver, as previously described (8). The incubation mixture contained 12.0 mμmoles each of twenty amino acids per milliliter, including either DL-leucine-1-^{14}C (6.5 mC/mmole), DL-lycine-^{14}C (240 mC/mmole), or DL-valine-1-^{14}C (11 mC/mmole). One mixture contained ^{14}C-algal hydrolysate (1.8 mC/mg) instead. The ^{14}C-aminoacyl-RNA was isolated and dialyzed for 6 hours at 4° against 0.01 M acetate (K^+), pH 5.1, containing 0.05 M KCl. The preparation was passed through a 2 × 35-cm column of Sephadex G-25 equilibrated with the same buffer. The absorbance at 260 mμ of the band of RNA eluted initially from the column was determined. The RNA solution was diluted to 3 mg per milliliter, assuming 1 mg of RNA per milliliter has an absorbance of 23. The preparation was stored at −60°.

TRANSFER SYSTEM ASSAY

Transfer of amino acids from amino acyl-RNA to ribosome-bound polypeptides was studied in a mixture containing the following components in a total volume of 1.5 ml: 0.1 μmole GTP, 5 μmoles $MgCl_2$, 100 μmoles KCl, 50 μmoles Tris-KCl, pH 7.5, at 23°, 15 μmoles GSH, 0.1 μmole each of 20 amino acids, 300 μg each of enzyme fractions TF-1 and TF-2, and amounts of KCl-washed ribosomes and ^{14}C-aminoacyl-RNAs as specified in the legends to the figures and tables. The mixture was incubated for 3 minutes at 37°. The assays were chilled, and casein was added as carrier to give a total protein weight of approximately 15 mg. The protein was precipitated with 5% trichloroacetic acid, washed, and assayed for radioactivity by liquid scintillation counting, as described previously (9). Incubation mixtures which contained crude enzyme preparations with substantial amounts of hemoglobin were precipitated first with acidified acetone (0.2 ml of 12 N HCl/100 ml acetone). This procedure removed the heme groups which otherwise would quench the radioactivity during liquid scintillation counting.

ANALYSES OF RIBOSOME-BOUND PRODUCTS

After some of the incubations the labeled ribosomes were isolated by centrifugation and precipitated with cold 5% trichloroacetic acid. The precipitate was washed twice with cold acid and dissolved in 1 N NaOH. Upon reacidification some of the ^{14}C-products remained soluble, while the others precipitated. Both acid-soluble and acid-insoluble products were analyzed for radioactivity. In some experiments it was desirable to determine the percentage of ^{14}C-radioactivity in the acid-soluble product which was located in free amino acid molecules. The reaction mixture was incubated initially with valine-1-^{14}C, and the labeled ribosomes were isolated. The acid-soluble product obtained from these ribosomes was treated with chloramine T to decarboxylate the free amino acids, and the liberated CO_2 was trapped for analysis of radioactivity (10).

RESULTS

Ribosome and enzyme preparations. Enzyme activity for transferring amino acid from aminoacyl-RNA to polypeptide was measured in a reaction mixture containing ribosomes washed with a KCl solution (see METHODS). Table I shows that addition of a crude enzyme fraction to a system containing unwashed ribosomes provided only a

TABLE I

EFFECT OF WASHING THE RIBOSOMES ON REQUIREMENT FOR SOLUBLE ENZYME IN TRANSFER ASSAY[a]

Treatment of ribosomes	Incorporation ($\mu\mu$moles)	
	Without enzyme	With enzyme
Unwashed	141	235
Washed without KCl[b]	63	200
Washed with KCl[c]	3	185

[a] The transfer assay contained 3 mg of ribosomes and 300 μg of leucyl-^{14}C–RNA (475 $\mu\mu$moles of leucine-^{14}C); 3 mg of the ammonium sulfate fraction (see METHODS) were used as enzyme.

[b] Ribosomes obtained by a single centrifugation (unwashed ribosomes) were suspended in medium B (see METHODS) and reisolated by centrifugation.

[c] Ribosomes obtained by a single centrifugation were suspended in medium B containing 0.1 M KCl (see METHODS) and reisolated by centrifugation.

twofold increase in transfer of leucine-^{14}C into polypeptide. The ribosomes had substantial amounts of transfer enzyme activity associated with them (Table I). Suspension of the unwashed ribosomes in medium B (see METHODS) followed by centrifugation removed much of this enzyme activity (Table I). However, other experiments revealed that these ribosomes were still inadequate for demonstrating a requirement for both of the purified enzymes TF-1 and TF-2 in a transfer assay. Significant amounts of the TF-2 fraction were bound very tightly to the ribosomes. Washing the ribosomes with 0.10 M KCl in medium B removed this remaining enzyme activity. A transfer reaction using these ribosomes now required added enzyme fractions (Table I). Others have reported that crude ribosome fractions isolated from *Escherichia coli* and rat liver have transfer enzyme activities which can be removed by washing with salts or desoxycholate (11, 12).

The preparation of the two transfer fractions was modified from that briefly reported earlier (3) in order to improve the yield and separation. The procedure (Fig. 1) consists of a protamine sulfate treatment of the reticulocyte soluble phase to remove sRNA, an $(NH_4)_2SO_4$ fractionation, and a partial separation of the two enzyme activities using a calcium phosphate gel. Further purification of the two fractions was obtained by chromatography on DEAE-cellulose columns.

Fractionation of the 78,000*g* supernatant phase of the cell with $(NH_4)_2SO_4$ increased the enzyme specific activity for transfer about 12-fold (Table II). The ammonium sulfate fraction contained all the soluble enzymes needed to incorporate amino acids into hemoglobin, including the aminoacyl-RNA synthetases (5), but most of the hemoglobin in the cell lysate had been removed. Further purification using a calcium phosphate gel removed almost all of the remaining hemoglobin in the discarded S-1 fraction (Fig. 1). Gel fraction S-2 contained the peptide synthetase enzyme (TF-2) and was devoid of the binding enzyme (TF-1). Hence, the S-2 fraction alone could not catalyze transfer of leucine-^{14}C into polypeptides (Table II). Gel fraction S-3 contained the binding enzyme and significant amounts of the peptide synthetase; this crude fraction had a specific activity for transfer 30–40 times that of the 78,000*g* supernatant phase (Table II).

The specific activities of the purified fractions TF-1 and TF-2 were not measured in the experiments shown in Table II, because separately they did not stimulate transfer. However, other assays revealed that the TF-1 fraction catalyzed the transfer of about 400$\mu\mu$moles of leucine-^{14}C per milligram of enzyme protein when saturating amounts of the TF-2 fraction and 3 mg of ribosomes were used. Conversely, the TF-2 fraction catalyzed the transfer of about 600 $\mu\mu$moles of leucine-^{14}C per milligram of enzyme protein when the TF-1 fraction was present in excess.

Furthermore, the specific activity of the TF-1 fraction for the binding of phenylalanyl-RNA to ribosomes in the presence of poly U (2) was found to be 100-fold greater than that of the 78,000*g* supernatant phase for the total transfer of phenylalanine. In this poly U system the amount of phenylalanine transferred in the presence of both

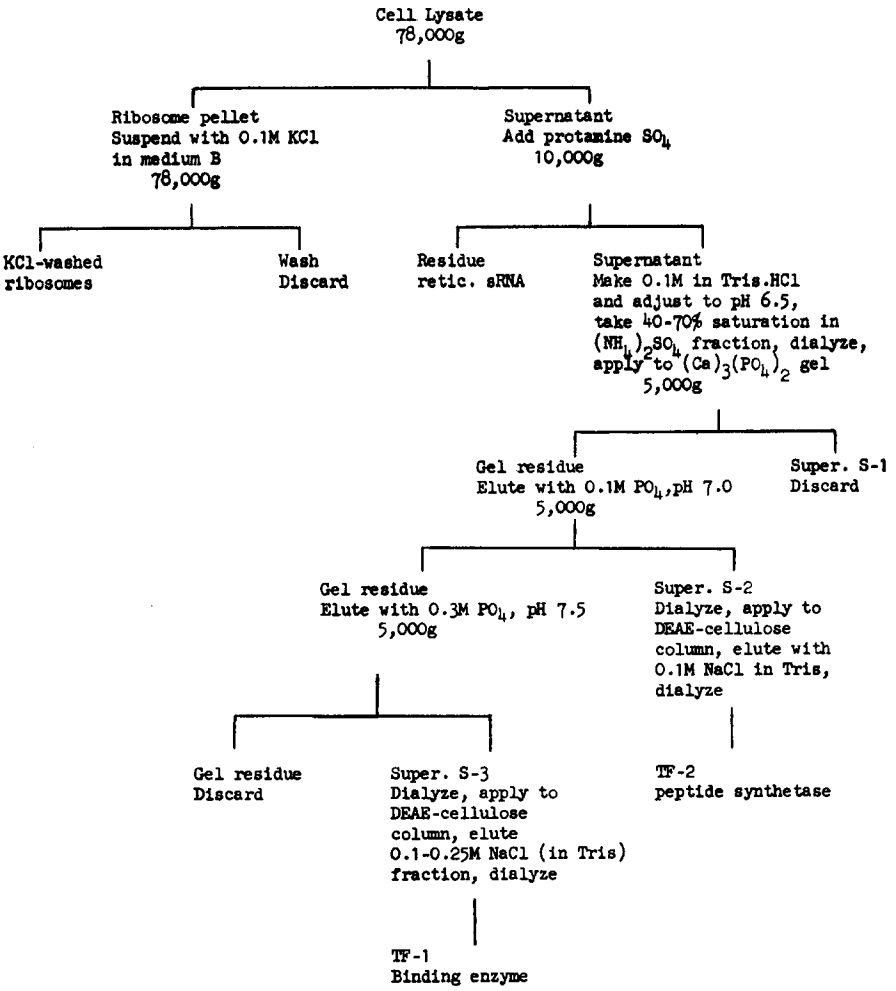


FIG. 1. Scheme for preparation of transfer enzymes and KCl-washed ribosomes.

fractions TF-1 and TF-2 was always greater than the amount of phenylalanyl-RNA bound to the ribosomes with TF-1 alone. Although the binding activity of the crude supernatant phase could not be measured directly because of the contaminating peptide synthetase, these results suggest that the binding activity was purified at least 100-fold in the preparation of the TF-1 fraction. Too, in other assays the TF-1 fraction did not contain phenylalanyl-RNA synthetase activity.

Requirements for the transfer reaction. Transfer of amino acids from aminoacyl-RNA into polypeptides required both enzyme fractions TF-1 and TF-2 (Fig. 2; Table III). The usual synergism was six to ten times the sum of the separate incorporations catalyzed by each fraction alone. When either enzyme fraction was heated briefly at 100° and then used with the complementary enzyme, there was no incorporation.

Table III shows that transfer of leucine-^{14}C into polypeptides required GSH. We had reported earlier that GSH was required to preserve enzyme activity for transfer in studies using a crude ammonium sulfate fraction (13). In the present studies this requirement was demonstrated only if the TF-2 fraction was freed of GSH and EDTA and stored frozen in their absence. There was still activity for transfer when the TF-2 fraction was freed of GSH and used immediately in an assay without GSH. Thus,

TABLE II

PURIFICATION OF TWO ENZYME FRACTIONS REQUIRED FOR TRANSFER OF AMINO ACIDS FROM AMINOACYL-RNA TO POLYPEPTIDES[a]

Enzyme fractions	Volume (ml)	Protein (mg)	Total activity (%)	Specific activity[b]
78,000g supernatant	1700	94,000	100	10.4
40–70% $(NH_4)_2SO_4$	188	3752	51	130.0
$(Ca)_3(PO_4)_2$ gel S-1	236	2030	6	26.0
$(Ca)_3(PO_4)_2$ gel S-2[c]	71	405	—	—
$(Ca)_3(PO_4)_2$ gel S-3	77	330	12	340.0
DEAE-cellulose TF-1[c]	25	100	—	—
DEAE-cellulose TF-2[c]	23	60	—	—

[a] Enzyme fractions were isolated as described in METHODS. The transfer assay (see METHODS) contained 3 mg of KCl-washed ribosomes, 300 μg of leucyl-^{14}C–RNA, and limiting amounts of the enzyme fractions.

[b] Specific activity is expressed as $\mu\mu$moles leucine-^{14}C transferred per milligram of enzyme protein.

[c] These fractions were inactive for transfer of leucine-^{14}C by themselves.

a sulfhydryl compound is needed in the transfer assay presumably to keep the TF-2 fraction in an active state. Hardesty and coworkers have reached a similar conclusion in their studies with a reticulocyte transfer system (14). Sutter and Moldave have reported also that GSH activates only one of the two transfer fractions in a system from rat liver (15).

The requirements for $MgCl_2$ (Table III) and KCl in the transfer reaction were almost absolute. Figure 3a shows that a concentration of 1.5–2.5 mM $MgCl_2$ was optimal for the transfer reaction. The optimal concentration of KCl was about 100 mM (Fig. 3b). These concentrations of $MgCl_2$ and KCl are very similar to those of the cell-free system incorporating free amino acids into hemoglobin (5). Hence, these data are consistent with the hypothesis that the transfer system is incorporating amino acids into globin polypeptides.

Table III also shows that GTP was required for the transfer of amino acids into polypeptides, as has been reported for other transfer systems. Maximum transfer required concentrations of 20 μM GTP or greater. Table IV shows that other nucleoside triphosphates and GDP were relatively inactive in stimulating transfer. Such specificity for GTP was found only if the aminoacyl-^{14}C–RNA preparations were passed through columns of Sephadex G-25 in order to remove small amounts of contaminating nucleoside triphosphates (see METHODS). In the presence of an optimal amount of GTP, addition of ATP, or creatine phosphate with creatine kinase did not increase incorporation.

About 25% of the radioactivity of the added aminoacyl-^{14}C–RNA was incorporated into polypeptides using the usual assay conditions (150 μg of aminoacyl-^{14}C–RNA and 1.5 mg of ribosomes per 1.5 ml). With greater amounts of aminoacyl-RNA the incorporation increased, while with lesser

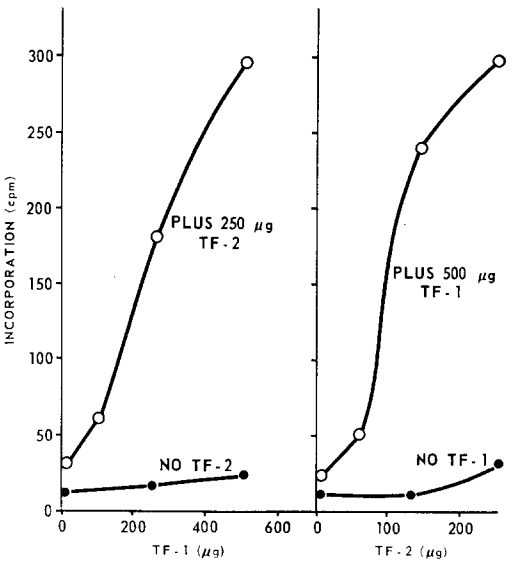


FIG. 2. Activity of two enzyme fractions in the transfer of leucine-^{14}C from leucyl-^{14}C–RNA into polypeptide. The transfer assay (see METHODS) contained 300 μg leucyl-^{14}C–RNA (650 $\mu\mu$moles of leucine-^{14}C) and 3 mg of ribosomes.

TABLE III

REQUIREMENTS FOR THE TRANSFER OF ^{14}C-AMINO ACIDS INTO POLYPEPTIDES[a]

Assay conditions	Incorporation ($\mu\mu$moles)
Complete	35.0
Minus TF-1 and TF-2	0.8
Minus TF-2	4.0
Minus TF-1	1.8
Minus GSH[b]	1.5
Minus GTP	0.4
Minus Mg^{2+}	0.3
Minus ribosomes	0.6

[a] The transfer assay contained the usual components (see METHODS) with 90 μg of leucyl-^{14}C–RNA (150 $\mu\mu$moles leucine-^{14}C) and 1.5 mg of ribosomes.

[b] The TF-2 enzyme fraction was passed through a Sephadex G-25 column equilibrated with 0.02 M Tris-HCl, pH 7.5, and stored frozen without GSH in order to inactivate the enzyme. The TF-2 fraction retained its activity when stored frozen with GSH (see text).

amounts of aminoacyl-RNA the total incorporation remained the same and the percentage of transfer increased. The transfer of ^{14}C-amino acid from aminoacyl-RNA to polypeptides reached a maximum after three minutes of incubation at 37°; no transfer occurred at 0°.

Ribosome-bound products. All synthesized ^{14}C-polypeptides were bound to the ribosomes when they were centrifuged from one of the transfer assays. This finding is attributed to the limiting amounts of aminoacyl-RNA used in the assays. Moreover, some additional radioactive products were bound to the isolated ribosomes. These products would not precipitate with trichloroacetic acid after the ribosome residue had been dissolved in NaOH (see METHODS). Such acid-soluble products normally were discarded during the acid washes used to

TABLE IV

EFFECT OF VARIOUS NUCLEOSIDE TRIPHOSPHATES ON TRANSFER[a]

Assay conditions	Incorporation ($\mu\mu$moles)
No nucleoside triphosphate	0.8
67 μM GTP	39.0
20 μM GTP	35.0
20 μM ATP	6.0
20 μM CTP	4.2
20 μM UTP	4.3
20 μM GDP	3.3

[a] The transfer assay contained the usual components (see METHODS) with 150 μg of lysyl-^{14}C–RNA (162 $\mu\mu$moles lysine-^{14}C) and 1.5 mg of ribosomes.

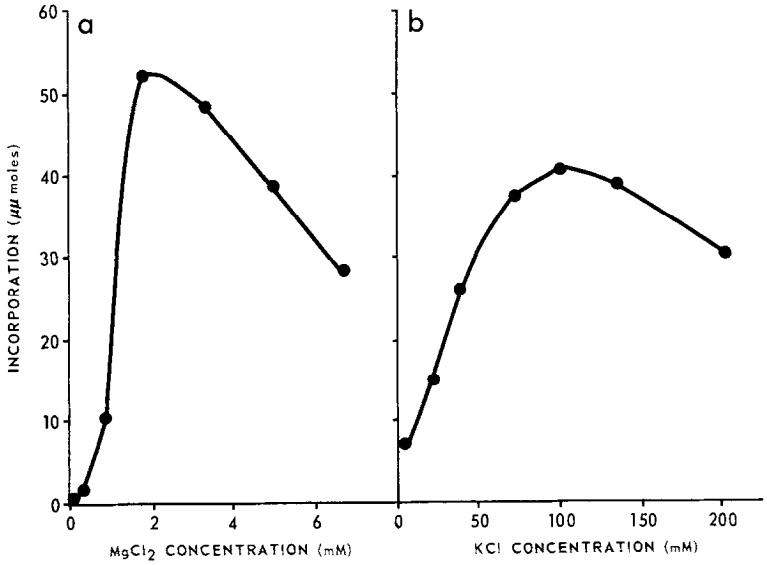


FIG. 3. Effect of varying the $MgCl_2$ and KCl concentrations on the transfer of lysyl-^{14}C–RNA into polypeptides. The transfer assay (see METHODS) contained 150 μg lysyl-^{14}C–RNA (160 $\mu\mu$moles of lysine-^{14}C) and 1.5 mg of ribosomes.

TABLE V

ANALYSIS OF RIBOSOME-BOUND PRODUCTS AFTER TREATMENT WITH NaOH[a]

Assay conditions	^{14}C-Amino acids in product ($\mu\mu$moles)	
	Acid-soluble	Acid-insoluble
Complete	61.5	139.4
Without TF-1 and TF-2	7.3	3.2
Without TF-2	13.2	15.9
Without TF-1	8.5	7.5

[a] The transfer assay contained 90 μg ^{14}C-amino-acyl-RNA (estimated 659 $\mu\mu$moles of algal hydrolyzate ^{14}C-amino acids) and 3 mg of ribosomes. After incubation the ribosomes were isolated by centrifugation, precipitated by 5% trichloroacetic acid, and dissolved in NaOH. The amounts of radioactivity which were soluble and insoluble upon subsequent acidification are given.

prepare ^{14}C-polypeptides for the analysis of radioactivity, because the wash procedure (5) included a step in which the residue was treated with NaOH. The acid-soluble products contained about 30% of the total radioactivity bound to the ribosomes (Table V). They are presumably amino acids and small peptides which were originally attached through ester bonds to sRNA molecules on the ribosomes.

A substantial percentage of the products of a poly U-stimulated transfer system were found previously to be phenylalanyl-RNA molecules bound to the ribosomes (2). It was of interest to know what percentage of the ribosome-bound radioactivity in the hemoglobin transfer system could be identified as amino acyl-RNA. For these experiments valyl-^{14}C–RNA containing valine labeled with ^{14}C in only the carbon atom of the carboxyl group was used. After a transfer incubation the labeled ribosomes were isolated by centrifugation. About 30% of the total radioactivity bound to the ribosomes was extracted in the acid-soluble fraction described in the preceding paragraph. These products were treated with chloramine T in a procedure which decarboxylates only free amino acids and collects the liberated-$^{14}CO_2$ (10). About 12% of the material was found to be free valine-^{14}C using this procedure, and the rest was presumably small peptides. Thus, of the total product bound to the ribosome, about 70% of the radioactivity was incorporated into polypeptides large enough to precipitate with trichloroacetic acid, 26% was in small peptides soluble in acid, and only 4% could be identified as amino acyl-RNA.

Although the exact nature of the acid soluble peptides is unknown, the percentage of radioactivity in this fraction remained the same when leucyl-^{14}C–RNA or phenylalanyl-^{14}C–RNA, instead of valyl-^{14}C–RNA, was used in the reaction. Therefore, it is doubtful that the small peptides represent newly started chains containing the first few amino acids coded by mRNA for hemoglobin, because there are no phenylalanine residues near the amino terminus of either rabbit hemoglobin chain (16, 17). Perhaps they represent an artifact phenomenon involving ribosomes isolated in the process of chain elongation which have lost a peptidyl-RNA during the preparative procedures; in this case, synthesis reinitiated in the cellfree system might produce some small labeled peptides coded by sections of mRNA other than those corresponding to the amino-terminal ends of the complete chains.

Only one of the transfer enzymes, the TF-1 fraction, is needed to stimulate the binding of phenylalanyl-RNA to ribosomes in the poly U transfer system (see DISCUSSION). An attempt was made to detect a similar reaction in the hemoglobin transfer system. However, Table V shows that substantial amounts of the acid-soluble products, including possible aminoacyl-RNA molecules, were found only when both transfer enzymes were used. Thus, in contrast to the poly U transfer system, the TF-1 fraction does not stimulate the binding of sizable amounts of aminoacyl-RNA to the ribosomes in the hemoglobin transfer system.

DISCUSSION

We have reported previously that the TF-1 and TF-2 enzyme fractions stimulate the formation of polyphenylalanine from phenylalanyl-RNA in the presence of poly U and ribosomes devoid of hemoglobin mRNA (2). In this poly U system the TF-1 fraction and GTP stimulated the binding of phenylalanyl-RNA to the ribosomes, while the TF-2 fraction catalyzed the formation of the peptide bond of phenylalanylphenylalanine,

in isolated reactions. Thus, these enzymes catalyze at least two sequental steps in polypeptide synthesis. When both enzymes were used in the same reaction mixture only 20% of the ribosome-bound radioactivity was located in polyphenylalanine molecules having a sufficient number of residues to be precipitable with trichloroacetic acid after NaOH treatment. The majority of the ribosome-bound products were oligopeptides too small to precipitate with acid. We suggest that the ability of the TF-1 fraction alone to stimulate the binding of phenylalanyl-RNA to the ribosomes and the preponderance of small peptides formed in the presence of both enzymes are manifestations of a system in which polypeptide chains are being initiated.

Conversely, in the transfer system documented in this report, about 70% of the ribosome-bound radioactivity was precipitable with trichloroacetic acid after NaOH treatment. This finding is consistent with the hypothesis that amino acids were incorporated into polypeptide chains which were interrupted in growth by cell lysis. We can only speculate about the reason for failure of one enzyme fraction with GTP to stimulate the binding of aminoacyl-RNA in this hemoglobin transfer system. Probably only a few globin chains, if any, are initiated in the reaction incubation. However, during the growth of existing polypeptide chains, the ribosome is thought to have at least two binding sites, one for peptidyl-RNA and one for the aminoacyl-RNA bearing the amino acid to be added to the growing chain (18, 19). If both of these sites are already occupied on the ribosomes added to the reaction mixture, it may be necessary for the peptide synthetase (TF-2) to catalyze peptide bond formation and the TF-1 fraction to catalyze a translocation of the new peptidyl-RNA (20) before a site is available for the binding of aminoacyl-RNA. In this event GTP and the TF-1 fraction presumably catalyze a reaction other than the physical binding of phenylalanyl-RNA, as discussed in the following paper (21). In the poly U system, phenylalanyl-RNA can bind to ribosomes containing chains of polyphenylalanine, but the available sites are thought to be "primed" by the prior incubation of these ribosomes with the transfer enzymes (20). Conversely, in the present transfer system a binding site may be available, but the complex formed is unstable to isolation by centrifugation.

Our studies with the two types of transfer systems using reticulocyte ribosomes have demonstrated that conventional methods of product preparation for analysis of incorporated radioactivity may give misleading results. These methods include a step in which the acid-precipitated protein is heated at 90° (22) or suspended in NaOH (5). These procedures remove the radioactivity in aminoacyl-RNA molecules from the polypeptide residue. Our work shows, however, that not only amino acids, but also those radioactive peptides too small to precipitate with acid, are removed. Thus, investigations studying chain initiation may not be able to use these conventional techniques to determine formation of the first peptide bond(s).

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